

Nikhil Rajan

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EDUCATION

London School of Economics and Political Science

London, UK

MSc in Financial Statistics

Sep 2026 - Jun 2027

- **Coursework:** Stochastic Processes, Mathematics of Black & Scholes Theory, Fixed Income, Interest Rates & Credit Risk, Stochastic Simulation and Calibration, Financial Statistics I & II (Machine Learning), Advanced Statistics.

Madras School of Economics

Chennai, India

Bachelor of Arts (Honours) in Economics

Aug 2023 - Jun 2026

- **Cumulative Grade Point Average:** 9.61/10.0 | **Gold Medallist (Rank 1/70).**
- **Dean's List (Distinction):** Semesters II - VI.
- **Coursework:** [Stochastic Processes](#), Real Analysis, [Advanced Time Series](#), Differential Equations, Linear Algebra, Econometrics, Statistics, Game Theory, Money and Banking, Data Science, Python and MATLAB.

Princeton University

Online

Macro, Money & Finance Summer School

May 2026 - Jul 2026

- Completed the Markus' Academy Ph.D.-level summer program run by Prof. Markus Brunnermeier, covering continuous-time macro-finance, stochastic calculus, and dynamic asset pricing.

QUANTITATIVE RESEARCH PROJECTS

Credit Rating Transitions and Markov Chains

2025

Reconstructed the Jarrow-Lando-Turnbull (1997) credit migration model from first principles and derived hedging strategies and risk-neutral transition matrices.

Hidden Markov Model Framework for Credit Stress Detection and Recession Early Warning

2026 (Ongoing)

Developing a hidden Markov model and mean-reverting process to identify credit stress regimes, assess their link with recessions, and develop a backtested tactical allocation strategy.

Liquidity Regimes as a Markov Chain and Optimal Execution as a Markov Decision Process

2025 (Ongoing)

Modelling liquidity as a four-state Markov process and exploring continuous-time optimal execution under persistent liquidity regimes.

Causal Analysis of Jump Contagion in Equity Markets: Evidence from Nifty 50 and S&P 500 Returns

2026

Examined jump spillovers between S&P 500 and Nifty 50 returns, finding a 0.284 transmission coefficient and $3.42\times$ higher volatility during crisis periods.

Stochastic Welfare Analysis of Sin Taxation Under India's New GST Regime

2024

Demonstrated that cigarette tax hikes increase household welfare in two Indian states using Cobb-Douglas utility and Markov chains.

TEACHING ASSISTANTSHIPS

Stochastic Processes

Jul 2025 - Dec 2025

Created detailed notes on martingales, Markov Decision Processes, and Brownian motion; led problem-solving sessions.

Foundations of Modern Finance I & II

May 2025 - Jul 2025

Conducted quizzes and weekly assignments; explained CAPM, Stochastic Discount Factors, and derivatives.

Time Series Analysis

Jan 2026 - May 2026

Led weekly problem-solving sessions on ARMA, ARIMA, VAR models, stationarity, forecasting, MLE, and Box-Jenkins modelling.

Game Theory

Jan 2026 - May 2026

Facilitated weekly problem-solving sessions on strategic-form games, Nash equilibrium, mixed strategies, and sequential games.

Multivariable Calculus

Jan 2025 - May 2025

Taught revision lectures and tutorials on sequences, series, gradients, partial derivatives, and Lagrange multipliers.

Intermediate Microeconomics

Jul 2025 - Dec 2025

Delivered classes and prepared assignments on monopoly pricing, exchange economies, and externalities using marginal analysis.

INTERNSHIP EXPERIENCE

Madras School of Economics

Chennai, India

Research Intern (Advisor: Dr. Naveen Srinivasan)

Dec 2024 - Jan 2025

- Investigated horizontal innovation and knowledge spillovers in the Romer growth model using analytical and numerical methods.
- Proposed and evaluated approaches to address a research question linking institutional quality, innovation, and productivity.

Yosaney Fintech Private Limited

Chennai, India

Finance Intern

May 2024 - Jul 2024

- Built an automated tool generating P&L, balance sheets and financial schedules, speeding reporting for 100+ MSMEs.
- Analysed MSME financial statements and evaluated bank/NBFC credit products for documentation and risk requirements.

EXTRACURRICULAR ACHIEVEMENTS

Chess

- Placed 13th on merit list in 1st SRM International Open FIDE Rapid Chess Tournament (2022).
- Captained the school team in CBSE Cluster Championships; scored 5.5/6 on Board 1 (2022).
- Placed 2nd in Under-15 category in Talent Chess Academy's 2nd FIDE-rated tournament (2019).

SKILLS

- **Quantitative & Financial Modelling:** Asset Pricing, Credit Risk Modelling, Stochastic Processes, Econometrics, Time Series, Market Microstructure, Game Theory.
- **Technical:** Python, MATLAB, Stata, Excel, LaTeX.
- **Analytical Foundations:** Real Analysis, Statistics, Differential Equations, Linear Algebra.